

Python Programming

Unit 01 – Lecture 05: Decision Making and Looping Structures

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Repository: <https://github.com/tali7c/Python-Programming>

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Learning Outcomes

- Write decisions using `if/elif/else` and nested conditions

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- Use `match-case` for multi-way branching (Python 3.10+)

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- Write decisions using `if/elif/else` and nested conditions
- Use `match-case` for multi-way branching (Python 3.10+)
- Write `for` and `while` loops using `range()`
- Control loops using `break`, `continue`, `pass`, and `loop else`

Decision Making: if/elif/else

```
marks = int(input("Marks: "))  
if marks >= 90:  
    print("A+")  
elif marks >= 75:  
    print("A")  
else:  
    print("Needs improvement")
```


Nested if (Example)

```
age = int(input("Age: "))
if age >= 18:
    if age >= 60:
        print("Senior")
    else:
        print("Adult")
else:
    print("Minor")
```

match-case (Python 3.10+)

- Useful for clean multi-way choices

```
day = int(input("Enter day number (1-7): "))
match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
    case _:
        print("Other day")
```

match-case (Python 3.10+)

- Useful for clean multi-way choices
- Alternative to long `elif` ladders

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Loops: for with range

- `range(n)` generates `0..n-1`

```
for i in range(1, 6):  
    print(i)
```

Loops: for with range

- `range(n)` generates `0..n-1`
- `range(start, stop, step)` is flexible

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for i in range(1, 6):  
    print(i)
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Loops: while

- Use when you do not know the exact number of iterations

```
n = 5
while n > 0:
    print(n)
    n -= 1
```

Loops: while

- Use when you do not know the exact number of iterations
- Always ensure the loop condition eventually becomes false

```
n = 5
while n > 0:
    print(n)
    n -= 1
```

Loop Control

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- `break`: stop the loop
- `continue`: skip to next iteration
- `pass`: do nothing (placeholder)
- `else in loops`: runs only if loop ends normally (no `break`)

else with a for Loop

```
for i in range(1, 6):  
    if i == 3:  
        break  
else:  
    print("Loop finished without break")
```

- Here else does not run because we used break.

Demo: Menu-Driven Program

- File: `demo/menu_driven_number_analyzer.py`

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- File: `demo/menu_driven_number_analyzer.py`
- Uses a `while True` loop + `match-case`
- Demonstrates clean branching + repetition until user exits

Checkpoint 1

Question: When does the else part of a loop execute?

Checkpoint 2

Question: What is the difference between `break` and `continue`?

Think-Pair-Share

Choose the better loop and justify:

- Printing 1..100
- Repeating until the user enters “exit”

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- Use `if/elif/else` for decisions and `match-case` for clean multi-way choices
- Use `for` when you have a count/range; use `while` for condition-based repetition
- `break` stops a loop; `continue` skips to next iteration
- Loop `else` runs only when no `break` happens

Exit Question

Write a loop that prints numbers 1..10 but skips multiples of 3.